NCUK INTERNATIONAL FOUNDATION YEAR

IFY Computer Science

Portfolio assessment

Note:
The total marks for this coursework are 30.

Instructions to Students

- You should answer these questions individually after the practice exercises for the topic you are studying. You can ask your tutor for advice if you have difficulty.
- You should code your solutions into an IDE and screenshot your program and the outputs from testing.
- You should paste your answers to the portfolio tasks in the NCUK portfolio answer template and upload them to the VLE.
    - For each program, you should include the code and a screen shot of the output.
    - The first three lines of a program should be comments consisting of
        - # Your Name
        - # Portfolio exercise x, where x is the portfolio question number.
        - # Brief description of program purpose
    - You should save the program in a file called P1.py, P2.py etc
- You should complete the exercises during the week(s) indicated and submit your portfolio for assessment and feedback according to your tutor's instructions.

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Grading Criteria

Each portfolio exercise is worth FIVE (5) marks. Exercises become progressively more difficult and complex throughout the course. Each exercise is marked out of 5: 2 for design and 3 for functionality.

Design marks are awarded for the approach you take to structuring the solution. This includes using meaningful variable names, comments, logical structure, and evidence of algorithmic thinking.

Functionality marks are awarded based on how well the program runs in line with the task requirements. **You can only gain functionality marks where there is clear evidence of a range of testing.**

Mark	**D1**		**D2**
Design Criteria	There is some evidence of planning or design in the code. Variable names are present but may be inconsistent or unclear. The program structure may be basic or partially correct, but the overall design shows a limited or incomplete approach. There may be signs of an intended solution, but it is underdeveloped.		The program shows a clear and well-considered design. Variable names are meaningful and consistent, the logic is easy to follow, and there is evidence of modular thinking and good programming techniques. The approach is appropriate to the task and shows planning, even if the final program does not fully work.
Mark	**F1**	**F2**	**F3**
Functionality Criteria	The program is partially functional. There is evidence of a working solution, but some errors or omissions prevent it from meeting all task requirements.	The program runs without syntax or runtime errors. All key elements of the task are implemented correctly, and the solution is complete and functional.	The program is **fully** functional, accurate, and complete. It runs without errors and meets all task requirements. In addition, it includes enhanced or extended functionality beyond the scope of the original task. This may include additional features that demonstrate independent thinking and advanced technical skill.

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Exercise	Topic, Focus & SLOs	Portfolio activities
P1	**SLO4** **AC4.1** **Input & Output /** **math Module**	**P1 Stadium Screens** **Your Task:** A sports stadium has made two spaces available and hopes to install two new rectangular video screens. The stadium has: • Space for one **large screen**, with a maximum diagonal length of **12 metres**. • Space for one **small screen**, with a maximum diagonal length of **7.52 metres**. The larger suitable screen should always be allocated to the large-screen space. A second screen can only be installed if its diagonal does not exceed 7.52 metres. Write a Python program that allows stadium staff to enter the dimensions of two proposed screens and determine whether each screen is suitable for installation. Your program must: • Input the height and width of both proposed screens. • Calculate the diagonal length of each screen using Pythagoras' theorem. • Determine whether each screen fits in one of the available spaces. • Display how many screens will fit and clearly identify which screen or screens are suitable. • Display clear and helpful messages for the stadium staff. Your program must also: • Use clear and meaningful variable names. • Include comments explaining each main part of the program.

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- Follow good formatting practices, including consistent indentation and spacing.

A diagonal can be calculated using:

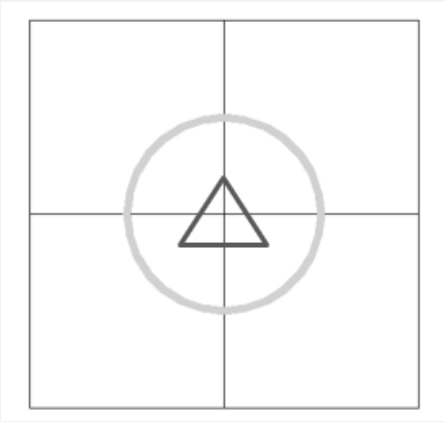
$$\text{diagonal} = \sqrt{\text{height}^2 + \text{width}^2}$$

Test Data

You should use the data below to test your solution.
You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected.

Test	Screen 1 dimensions	Screen 2 dimensions	Expected result
1	Height: 15 m, Width: 8 m	Height: 10.5 m, Width: 5.6 m	One screen fits: Screen 2 fits in the large-screen space
2	Height: 10 m, Width: 6 m	Height: 7 m, Width: 2.5 m	Both screens fit
3	Height: 13 m, Width: 5 m	Height: 8 m, Width: 4 m	Neither screen fits

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P2	**SLO4** **AC4.1** **Modules**	**P2 Logo Creation** A design company has asked you to create a logo using Python's turtle module. Your program must reproduce the design shown below. • A circle and outside rectangle must be centred on the horizontal and vertical grid lines. • The triangle may be positioned anywhere inside the circle, but one vertex must point upwards. • The outside rectangle is 360 pixels wide and 300 pixels high. • The circle has a diameter of 160 pixels. • The triangle can be any size but must fit fully inside the circle. • The circle's outline is coloured blue, the triangle's outline is coloured green, and all other lines are coloured red. • No shape is filled in. 

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P3	**SLO4** **AC4.1** **Convert pseudocode to Python code**	**P3 Weekly Temperature** A student has written an algorithm represented using pseudocode. The algorithm should: • ask the user to enter a temperature for each of five days • output the number of days that the temperature is greater than 15 • output the number of days that the temperature is less than or equal to 15 The algorithm **does not work** as expected. 1. Set both high and low to zero. 2. Store the text "5" in days. 3. Display the message: **"Enter 5 values."** 4. Convert "5" from a string into the integer 5 and store it in a new variable. 5. Repeat the following steps five times: a. Ask the user to enter a temperature. b. If the temperature is greater than 15, add 1 to high. c. Add 1 to low. 6. Display the final values of high and low with a suitable message. Create a Python program based on this design. At the end of your code, add a comment that identifies the error in the student's design and explains how you corrected it. **Test Data** You should use the data below to test your solution.

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		You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected. **Test** | **Temperatures** | **Expected result** 1 | 16, 17, 18, 19, 20 | Above 15: 5; 15 or below: 0 2 | 10, 11, 12, 13, 14 | Above 15: 0; 15 or below: 5 3 | 10, 15, 16, 20, 8 | Above 15: 2; 15 or below: 3

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P4	**SLO4** **AC4.1** **Selection** **String Concatenation**	**P4 Comic Con Entry Ticket** **Your Task** A Comic Con events company needs a program to calculate the price of a customer's entry ticket. The ticket price depends on the customer's age and whether they are a club member. Use the information in the table below to determine the ticket price. **Customer category** / **Ticket price** Member — £25 Younger than 18 or older than 60 — £40 Aged between 18 and 22 inclusive — £45 All other customers — £50 Each ticket is printed with a customer reference code which is made up of the first three letters of their surname, their age and the first letter of their first name. All letters in the reference code must be uppercase. *For example:* A non-member called Jamie Wilson, aged 20, would have the reference code WIL20J and would pay £45. A member called Samina Khan, aged 45, would have the reference code KHA45S and would pay £25. If a customer meets more than one condition, the program should apply the **first suitable category shown in the table**. For example, a member should pay £25 regardless of their age. Write a program that correctly calculates the ticket price and customer reference code. The program must produce a clearly formatted ticket that can be printed for the customer and used to gain entry to the event. Your program must: • Use clear and meaningful variable names. • Include comments explaining the main parts of the program. • Use appropriate string-manipulation methods. • Follow good formatting practices.

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		• Produce clear and helpful output. **Test Data** You should use the data below to test your solution. You must carry out at least the tests shown below and provide evidence of the results to check your solution works as expected.

Test	Name	Age	Member	Expected price	Expected reference
1	Alex Morgan	35	Y	£25	MOR35A
2	Sophie Khan	16	N	£40	KHA16S
3	Jamie Wilson	20	N	£45	WIL20J
4	Peter Jones	45	N	£50	JON45P

P5	**SLO4** **AC4.1** **Loops** **Mathematical Operators**	**P5 Binary Conversion Program** A program converts short binary patterns into a non-negative denary integer. A valid input must contain at most four binary digits. Each digit must be either 0 or 1. If the user enters an empty pattern, the program must end immediately. The program does not convert patterns longer than 4 bits and should provide the user with a suitable error message asking them to re-enter with a valid pattern. Create a function that checks whether every digit is either 0 or 1. The function should be used in your main program before any conversions. If all digits are valid, a multiplier is used to represent each binary place value, doubling after each digit is processed. • When the program encounters a 1, the corresponding place value is added to a running total. • When it encounters a 0, nothing is added.

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- Once all the digits have been processed, the denary result is displayed.

Your program must:

- Use clear and meaningful variable names.
- Include comments explaining the main parts of the program.
- Use a function to check that all digits are either 0 or 1
- Produce clear and helpful error messaging.

Test Data

You should use the data below to test your solution.
You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected.

Input	Expected output
1100	Displays 12
0000	Displays 0
11111	Displays and explains error
1012	Displays and explains error
Empty input	Program exits

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P6	**SLO4** **AC4.1** **File Handling** **Standard Algorithms**	**P6 Zoo Animal Search** A zoo stores the English and Latin names of its animals in a text file. Each line contains an animal's English name and its corresponding Latin name. The structure of each line is: ENGLISH NAME, LATIN NAME The data file is provided. Download and save a local copy named Task P6 Zoo Database.txt. You do not need to edit or add records to this file. **Your Task** Create a Python program that searches the text file, to determine whether a particular animal is listed in the zoo's records. The program must: • Read the animal records from the supplied text file. • Allow the user to search using either an English name or a Latin name. • Make the search case-insensitive, so entries such as 'African lion', 'African Lion' and 'african lion' are treated as the same search. • Confirm whether the animal is listed in the zoo's records. • If a match is found, display both the English and Latin names. • Display a clear message if the animal is not found. • Allow the user to carry out another search without restarting the program. **Search Algorithm** You may choose the search algorithm used by your program. In comments at the end of your code, you should: • Identify the algorithm you have used. • Justify why it is suitable for the supplied data.

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Test Data

You should use the supplied data file to test your solution.
You must carry out at least the tests shown below and provide evidence of the results to show your solution works as expected.

Test purpose	Search input	Expected result
Valid English name search	RED PANDA	Displays RED PANDA — AILURUS FULGENS
Valid Latin name search	PANTHERA LEO	Displays AFRICAN LION — PANTHERA LEO
Different case check	Giant panda	Displays GIANT PANDA — AILUROPODA MELANOLEUCA
Animal not listed	Domestic Cat	Displays a suitable “not found” message

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